K-theory of the Roe algebra

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This is the working note of an ongoing project and also expository notes from *Higher Index Theory*. We try to keep the note somewhat self-contained.

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1 Basic definitions

We work with proper metric spaces when discussing Roe algebras, and with locally compact, second countable, Hausdorff spaces when discussing geometric modules and localisation algebras. Roe and localised Roe algebras are defined using geometric modules; after passing to K-theory, the constructions are independent of the ample module up to the canonical identifications used in [WY20, Chs. 5–7].

1.1 Geometric modules, support, and propagation

Definition 1.1 ([WY20, Def. 4.1.1])**.** Let X be a locally compact, second countable, Hausdorff space. A *geometric X module* is a separable Hilbert space H_X equipped with a non-degenerate representation

$$\rho : C_0(X) \longrightarrow B(H_X).$$

It is *ample* if no non-zero element of $C_0(X)$ acts as a compact operator, and if H_X is infinite-dimensional.

Definition 1.2 ([WY20, Defs. 4.1.7 and 4.1.8])**.** Let H_X and H_Y be geometric modules over locally compact spaces X and Y, and let $T : H_X \to H_Y$ be bounded. The *support* of T, denoted

$$\mathrm{supp}(T) \subseteq Y \times X,$$

is the set of all pairs (y, x) such that

$$\chi_V T \chi_U \neq 0$$

for every pair of open neighbourhoods $V \ni y$ and $U \ni x$. If $X = Y$ is metric and $T \in B(H_X)$, the *propagation* of T is

$$\mathrm{prop}(T) := \sup\{d(y, x) : (y, x) \in \mathrm{supp}(T)\} \in [0, \infty].$$

Definition 1.3 ([WY20, Def. 5.1.1])**.** Let H_X be a geometric module over a metric space X. A bounded operator $T \in B(H_X)$ is *locally compact* if, for every compact subset $K \subseteq X$,

$$\chi_K T \qquad \text{and} \qquad T\chi_K$$

are compact. It has *finite propagation* if $\mathrm{prop}(T) < \infty$.

Definition 1.4 ([WY20, Def. 5.1.4])**.** The *Roe $*$-algebra* of H_X is

$$C[H_X] := \{T \in B(H_X) : T \text{ is locally compact and has finite propagation}\}.$$

The *Roe algebra* of H_X is the norm closure of $C[H_X]$ in $B(H_X)$, and is denoted

$$C^*(H_X).$$

If we have fixed an ample module over X, we write simply $C^*(X)$.

We will use $K_0(A)$ for stable classes of projections or idempotents, and $K_1(A)$ for stable classes of unitaries or invertibles, with the usual unitisation convention in the non-unital case and with the conventions from [WY20, Ch. 2].

1.2 Localisation and localised Roe algebras

Definition 1.5 ([WY20, Defs. 6.2.1 and 6.2.3])**.** Let H_X be an X module. Define $L[H_X]$ to be the $*$-algebra of all bounded functions

$$(T_t)_{t\in[1,\infty)} : [1,\infty) \longrightarrow B(H_X)$$

such that:

(i) for every compact $K \subseteq X$, there is $t_K \geq 0$ such that, for all $t \geq t_K$,

$$\chi_K T_t \qquad \text{and} \qquad T_t \chi_K$$

are compact, and the functions

$$t \longmapsto \chi_K T_t, \qquad t \longmapsto T_t \chi_K$$

are uniformly norm continuous on $[t_K, \infty)$;

(ii) for every open neighbourhood U of the diagonal in $X^+ \times X^+$, there is $t_U \geq 1$ such that, for all $t > t_U$,

$$\operatorname{supp}(T_t) \subseteq U.$$

The *localisation algebra* $L^*(H_X)$ is the C^*-algebra completion of $L[H_X]$ for the supremum norm

$$\|(T_t)\| := \sup_{t\in[1,\infty)} \|T_t\|_{B(H_X)}.$$

If H_X is ample, we write $L^*(X) := L^*(H_X)$.

Definition 1.6 ([WY20, Def. 6.3.1])**.** If H_X is ample, the K-homology groups of X are

$$K_n(X) := K_n(L^*(X)), \qquad n \geq 0.$$

Thus $K_*(X) = K_0(X) \oplus K_1(X)$.

Definition 1.7 ([WY20, Def. 6.6.1])**.** Let H_X be a geometric module. The *localised Roe $*$-algebra $C_L[H_X]$* is the $*$-algebra of all uniformly continuous, bounded functions

$$(T_t)_{t\in[1,\infty)} : [1,\infty) \longrightarrow C[H_X]$$

such that, for every open neighbourhood U of the diagonal in $X^+ \times X^+$, there is $t_U \geq 1$ such that, for all $t > t_U$,

$$\operatorname{supp}(T_t) \subseteq U.$$

The *localised Roe algebra* $C_L^*(H_X)$ is the completion of $C_L[H_X]$ in the supremum norm. For metric computations below, we also use the propagation-to-zero model; by [WY20, Exercise 6.8.15], the natural inclusion of that model into $C_L^*(H_X)$ induces an isomorphism on K-theory.

Thus (T_t) is a uniformly continuous family of Roe operators whose supports shrink to the diagonal. If we fix an ample module over X, we abbreviate $C^*(H_X)$ and $C_L^*(H_X)$ to $C^*(X)$ and $C_L^*(X)$.

Proposition 1.8 ([WY20, Prop. 6.6.2]). *If H_X is an ample geometric module, then the natural inclusion*

$$C_L^*(H_X) \longrightarrow L^*(H_X)$$

induces an isomorphism on K-theory. In particular,

$$K_*(C_L^*(H_X)) \cong K_*(X),$$

where the right-hand side is the analytic K-homology of X.

Lemma 1.9 ([WY20, Lem. 5.1.8]). *If X is bounded, then*

$$C^*(H_X) = \mathcal{K}(H_X)$$

for every X module H_X.

2 Relative Roe algebras and full-corner identifications

2.1 The relative algebras

Let X be a proper metric space and let $E \subseteq X$ be closed. Choose a countable dense set $Z \subseteq X$ adapted to the closed subsets under discussion. In particular, $Z \cap E$ is dense in E; when a cover $X = E \cup F$ is used below, choose countable dense subsets $Z_E \subseteq E$ and $Z_F \subseteq F$ with $Z_E \cap Z_F$ dense in $E \cap F$, and put $Z := Z_E \cup Z_F$, as in the proof of [WY20, Thm. 7.5.5]. Let H be a separable infinite-dimensional Hilbert space. Put $H_X := \ell^2(Z, H)$ and $H_E := \ell^2(Z \cap E, H) = \chi_E H_X$. We write χ_E both for the characteristic function of E and for the corresponding projection $H_X \to H_E$.

Definition 2.1 (cf. [WY20, proof of Thm. 7.5.5]). Define the C^*-subalgebra

$$C_X^*(E) := \overline{\operatorname{span}\{S\chi_E T : S, T \in C^*(X)\}} \subseteq C^*(X).$$

Equivalently, $C_X^*(E)$ is the ideal of $C^*(X)$ generated by the multiplier χ_E. Similarly,

$$C_{L,X}^*(E) := \overline{\operatorname{span}\{(S_t)\chi_E(T_t) : (S_t), (T_t) \in C_L^*(X)\}} \subseteq C_L^*(X)$$

is the ideal generated by the multiplier χ_E in the localised Roe algebra.

The algebra $C_X^*(E)$ can also be described as the closure of the finite-propagation, locally compact operators on H_X supported in some neighbourhood of E, i.e. operators T for which

$$\chi_{N_r(E)} T = T \chi_{N_r(E)} = T$$

for some $r > 0$, where $N_r(E) := \{x \in X : d(x, E) \le r\}$. The analogous pointwise description applies to the dense subalgebra of $C_{L,X}^*(E)$. We have analogous constructions for $F \subseteq X$ and $E \cap F \subseteq X$.

2.2 Extension by zero and full corners

Write

$$H_X = H_E \oplus H_E^{\perp}.$$

If $T \in C^*(E)$, its *extension by zero* to H_X is the operator

$$\widetilde{T} = \begin{pmatrix} T & 0 \\ 0 & 0 \end{pmatrix}.$$

Finite propagation for the subspace metric on E remains finite propagation in X, and local compactness is preserved under extension by zero.

Proposition 2.2 (cf. [WY20, Prop. 2.7.19 and proof of Thm. 7.5.5])**.** *The extension-by-zero map identifies $C^*(E)$ with the corner*

$$\chi_E C^*_X(E) \chi_E.$$

Moreover, this corner is full in $C^*_X(E)$. Consequently, the inclusion

$$C^*(E) \cong \chi_E C^*_X(E) \chi_E \hookrightarrow C^*_X(E)$$

induces an isomorphism on K-theory.

Proof. If $T \in C^*(E)$, then $\widetilde{T} \in C^*(X)$ and

$$\widetilde{T} = \chi_E \widetilde{T} \chi_E,$$

so the image lies in $\chi_E C^*_X(E) \chi_E$.

Conversely, let $a \in \chi_E C^*_X(E) \chi_E$. Since a vanishes on $H_E^\perp$ and preserves H_E, there exists a bounded operator $T : H_E \to H_E$ with

$$a = \begin{pmatrix} T & 0 \\ 0 & 0 \end{pmatrix}.$$

On the dense $*$-subalgebra generated by finite-propagation terms of the form $S\chi_E U$, compression to H_E produces finite-propagation, locally compact operators on H_E. Passing to the norm closure gives $T \in C^*(E)$.

For fullness, a dense generating set for $C^*_X(E)$ is spanned by terms $S\chi_E T$. But

$$S\chi_E T = (S\chi_E)\chi_E(\chi_E T) \in C^*_X(E)\chi_E C^*_X(E).$$

Thus the ideal generated by the corner projection χ_E contains a dense generating set for $C^*_X(E)$. The K-theory statement follows because inclusion of a full corner induces an isomorphism on K-theory [WY20, Prop. 2.7.19]. ◻

Proposition 2.3 (cf. [WY20, Prop. 2.7.19 and proof of Thm. 7.5.5])**.** *After fixing the extension-by-zero embedding, $C^*_L(E)$ identifies with the full corner*

$$\chi_E C^*_{L,X}(E) \chi_E \subseteq C^*_{L,X}(E).$$

Consequently, the inclusion

$$C^*_L(E) \cong \chi_E C^*_{L,X}(E) \chi_E \hookrightarrow C^*_{L,X}(E)$$

induces an isomorphism

$$K_*(C^*_L(E)) \cong K_*(C^*_{L,X}(E)).$$

The same statements hold with E replaced by F or $E \cap F$.

Proof. Apply the previous argument pointwise in the parameter t. If $(T_t) \in C_L^*(E)$, then extension by zero gives a family

$$\widetilde{T}_t = \begin{pmatrix} T_t & 0 \\ 0 & 0 \end{pmatrix}$$

that is uniformly continuous, locally compact, and has support shrinking to the diagonal in X; in the propagation-to-zero model, its propagation goes to zero. Conversely, compression to H_E recovers a family in $C_L^*(E)$, since neighbourhoods of the diagonal in $E^+ \times E^+$ are obtained by restricting neighbourhoods of the diagonal in $X^+ \times X^+$. For fullness, each generator $(S_t)\chi_E(T_t)$ belongs to $C_{L,X}^*(E)\chi_E C_{L,X}^*(E)$. The K-theory statement follows from [WY20, Prop. 2.7.19]. □

3 Coarsely excisive decompositions and Mayer–Vietoris

3.1 Coarsely excisive covers

Definition 3.1 ([WY20, Def. 7.5.3])**.** Let $X = E \cup F$ be a cover by closed subsets. The cover is *coarsely excisive* if, for every $r > 0$, there exists $s > 0$ such that

$$N_r(E) \cap N_r(F) \subseteq N_s(E \cap F).$$

Example 3.2 ([WY20, Example 7.5.4])**.** For

$$E = \mathbb{R}^{n-1} \times (-\infty, 0], \qquad F = \mathbb{R}^{n-1} \times [0, \infty),$$

we have $E \cap F = \mathbb{R}^{n-1} \times \{0\}$, and

$$N_r(E) \cap N_r(F) = \mathbb{R}^{n-1} \times [-r, r] \subseteq N_r(E \cap F).$$

So this cover is coarsely excisive.

3.2 Intersection and sum identities

Proposition 3.3 (cf. [WY20, proof of Thm. 7.5.5])**.** *If $X = E \cup F$ is coarsely excisive, then*

$$C_X^*(E) \cap C_X^*(F) = C_X^*(E \cap F), \qquad C_X^*(E) + C_X^*(F) = C^*(X),$$

and similarly

$$C_{L,X}^*(E) \cap C_{L,X}^*(F) = C_{L,X}^*(E \cap F), \qquad C_{L,X}^*(E) + C_{L,X}^*(F) = C_L^*(X).$$

Proof. The inclusion $C_X^*(E \cap F) \subseteq C_X^*(E) \cap C_X^*(F)$ is immediate. For the reverse inclusion, pass to the dense finite-propagation subalgebras used in the proof of [WY20, Thm. 7.5.5]. A finite-propagation operator lying in both relative dense subalgebras is supported near E and near F, hence near $N_r(E) \cap N_r(F)$ for some r. By coarse excision, this lies in a neighbourhood of $E \cap F$. Taking closures gives the reverse inclusion.

For the sum identity, use the multiplier decomposition $T = \chi_E T + (1 - \chi_E)T$. The first term lies in $C_X^*(E)$, and since $X = E \cup F$, the second term is supported over F and lies in $C_X^*(F)$; equivalently, we obtain these terms by applying approximate units in the corresponding ideals. Hence $C_X^*(E) + C_X^*(F) = C^*(X)$. The localised statements are the corresponding pointwise statements for localised Roe families; the shrinking support condition is preserved, and the proof of [WY20, Thm. 7.5.5] notes that this part is similar and uses closedness. ◻

3.3 The pushout cube and the six-term sequences

Because of proposition 3.3, the relative algebras fit into a commutative cube of pushout diagrams. The left- and right-hand vertices record both the original intersection/sum expressions and the simplifications from proposition 3.3.

$$\begin{tikzcd}
\substack{C^*_{L,X}(E)\cap C^*_{L,X}(F)\\ = C^*_{L,X}(E\cap F)} \arrow[rr] \arrow[dr, "e_V"] \arrow[dd] & & C^*_{L,X}(E) \arrow[dr, "e_V"] \arrow[dd] & \\
& \substack{C^*_X(E)\cap C^*_X(F)\\ = C^*_X(E\cap F)} \arrow[rr] \arrow[dd] & & C^*_X(E) \arrow[dd] \\
C^*_{L,X}(F) \arrow[rr] \arrow[dr, "e_V"] & & \substack{C^*_{L,X}(E)+C^*_{L,X}(F)\\ = C^*_L(X)} \arrow[dr, "e_V"] & \\
& C^*_X(F) \arrow[rr] & & \substack{C^*_X(E)+C^*_X(F)\\ = C^*(X)}
\end{tikzcd}$$

We apply [WY20, Prop. 2.7.15] once to the top pushout square and once to the bottom pushout square. The evaluation-at-one maps assemble these into a natural comparison between the localised and non-localised Mayer–Vietoris sequences. Then the full-corner identifications in propositions 2.2 and 2.3 replace the relative algebras by the Roe and localised Roe algebras of E, F, and $E\cap F$.

Theorem 3.4 ([WY20, Thm. 7.5.5])**.** *Let $X = E\cup F$ be a coarsely excisive decomposition of a proper metric space. Then there are six-term exact sequences*

$$\begin{tikzcd}
K_0(C^*(E\cap F)) \arrow[r] & K_0(C^*(E))\oplus K_0(C^*(F)) \arrow[r] & K_0(C^*(X)) \arrow[d, "\partial_0"] \\
K_1(C^*(X)) \arrow[u, "\partial_1"] & K_1(C^*(E))\oplus K_1(C^*(F)) \arrow[l] & K_1(C^*(E\cap F)) \arrow[l]
\end{tikzcd}$$

and

$$\begin{tikzcd}
K_0(C^*_L(E\cap F)) \arrow[r] & K_0(C^*_L(E))\oplus K_0(C^*_L(F)) \arrow[r] & K_0(C^*_L(X)) \arrow[d, "\partial_0^L"] \\
K_1(C^*_L(X)) \arrow[u, "\partial_1^L"] & K_1(C^*_L(E))\oplus K_1(C^*_L(F)) \arrow[l] & K_1(C^*_L(E\cap F)). \arrow[l]
\end{tikzcd}$$

Moreover, for each $i\in\{0,1\}$, the evaluation-at-one map gives a commutative strip

$$\begin{tikzcd}
K_i(C^*_L(E\cap F)) \arrow[r] \arrow[d, "\mathrm{ev}_*"] & K_i(C^*_L(E))\oplus K_i(C^*_L(F)) \arrow[r] \arrow[d, "\mathrm{ev}_*\oplus\mathrm{ev}_*"] & K_i(C^*_L(X)) \arrow[d, "\mathrm{ev}_*"] \\
K_i(C^*(E\cap F)) \arrow[r] & K_i(C^*(E))\oplus K_i(C^*(F)) \arrow[r] & K_i(C^*(X)).
\end{tikzcd}$$

Proof. Apply [WY20, Prop. 2.7.15] to the top pushout square of the cube and again to the bottom pushout square. This gives a commutative diagram of Mayer–Vietoris sequences for $C^*_{L,X}(E)$, $C^*_{L,X}(F)$, $C^*_{L,X}(E \cap F)$ and for $C^*_X(E)$, $C^*_X(F)$, $C^*_X(E \cap F)$.

The corner identifications in propositions 2.2 and 2.3 replace the relative algebras $C^*_X(E)$, $C^*_X(F)$, $C^*_X(E \cap F)$ and $C^*_{L,X}(E)$, $C^*_{L,X}(F)$, $C^*_{L,X}(E \cap F)$ by the Roe and localised Roe algebras of E, F, and $E \cap F$. After these identifications, the relative Mayer–Vietoris sequences become the two six-term sequences above, and naturality gives the commutative strip. □

4 The half-space Eilenberg swindle

Proposition 4.1 ([WY20, Prop. 7.5.2]). *Let $Y := X \times [0, \infty)$ with the product metric*

$$d((x, t), (x', t')) = \sqrt{d_X(x, x')^2 + |t - t'|^2}.$$

Then

$$K_*(C^*(Y)) = 0.$$

Proof. Choose an ample module of the form

$$H_Y = H_X \otimes L^2([0, \infty)).$$

For each $n \geq 1$, let $V_n : L^2([0, \infty)) \to L^2([0, \infty))$ be the right-shift isometry

$$(V_n u)(t) = \begin{cases} u(t - n), & t \geq n, \\ 0, & 0 \leq t < n. \end{cases}$$

Define

$$H_Y^\infty := \bigoplus_{n \geq 1} H_Y,$$

which is again an ample Y module, and let $W_n : H_Y \to H_Y^\infty$ denote the inclusion into the n-th summand. First define Φ on the dense Roe $*$-algebra $C[H_Y]$. For $T \in C[H_Y]$, put

$$\Phi(T) := \sum_{n \geq 1} W_n (1 \otimes V_n) T (1 \otimes V_n^*) W_n^*.$$

Each summand is a translate of T along the half-line. Hence a compact subset of Y meets only finitely many of these translates, so $\Phi(T)$ is locally compact. Since the ranges of the W_n are orthogonal, the sum defines a bounded operator and $\|\Phi(T)\| \leq \|T\|$. Finite propagation is preserved under conjugation by $1 \otimes V_n$, so $\Phi(T) \in C[H_Y^\infty]$. Thus Φ is a contractive $*$-homomorphism on $C[H_Y]$, and it extends by continuity to

$$\Phi : C^*(Y) \longrightarrow C^*(H_Y^\infty).$$

Let $\Psi : C^*(Y) \to C^*(H_Y^\infty)$ be the map that places an operator into the first summand,

$$\Psi(T) = W_1 T W_1^*.$$

For every $T \in C^*(Y)$, the elements

$$\begin{pmatrix} \Phi(T) & 0 \\ 0 & \Psi(T) \end{pmatrix}, \qquad \begin{pmatrix} \Phi(T) & 0 \\ 0 & 0 \end{pmatrix}$$

of $M_2(C^*(H_Y^\infty))$ are conjugate by an isometry in the multiplier algebra of $M_2(C^*(H_Y^\infty))$. By the isometry trick [WY20, Prop. 2.7.5],

$$\Phi_* + \Psi_* = \Phi_*$$

as maps

$$K_*(C^*(Y)) \longrightarrow K_*(C^*(H_Y^\infty)).$$

Thus $\Psi_* = 0$.

On the other hand, Ψ is induced by a covering isometry for the identity coarse map $Y \to Y$, so Ψ_* is an isomorphism by [WY20, Thm. 5.1.15]. Hence $K_*(C^*(Y)) = 0$. □

Corollary 4.2 ([WY20, Prop. 7.5.2])**.** *For the same $Y = X \times [0, \infty)$,*

$$K_*(C_L^*(Y)) = 0.$$

Consequently, the assembly map for Y is an isomorphism.

Proof. By [WY20, Prop. 6.4.14], the K-homology of $X \times [0, \infty)$ is zero. By proposition 1.8, this is canonically isomorphic to $K_*(C_L^*(Y))$. The assembly map is therefore an isomorphism because both its domain and codomain vanish. □

5 The Roe algebra K-theory of Euclidean space

5.1 The inductive step

For $n \geq 1$, let

$$E_n := \mathbb{R}^{n-1} \times (-\infty, 0], \qquad F_n := \mathbb{R}^{n-1} \times [0, \infty).$$

Then

$$\mathbb{R}^n = E_n \cup F_n, \qquad E_n \cap F_n \cong \mathbb{R}^{n-1},$$

and example 3.2 shows that this cover is coarsely excisive. Moreover,

$$E_n \cong \mathbb{R}^{n-1} \times [0, \infty), \qquad F_n \cong \mathbb{R}^{n-1} \times [0, \infty),$$

so proposition 4.1 and corollary 4.2 give

$$K_*(C^*(E_n)) = K_*(C^*(F_n)) = 0, \qquad K_*(C_L^*(E_n)) = K_*(C_L^*(F_n)) = 0.$$

Proposition 5.1 (cf. [WY20, Thm. 7.5.5])**.** *For every $n \geq 1$, the Mayer–Vietoris boundary maps give isomorphisms*

$$K_0(C^*(\mathbb{R}^n)) \cong K_1(C^*(\mathbb{R}^{n-1})), \qquad K_1(C^*(\mathbb{R}^n)) \cong K_0(C^*(\mathbb{R}^{n-1})).$$

The same statement holds with C^* replaced by C_L^*.

Proof. Apply theorem 3.4 to $\mathbb{R}^n = E_n \cup F_n$. Since the K-theory of E_n and F_n vanishes, the six-term exact sequence collapses to

$$\begin{tikzcd} K_0(C^*(\mathbb{R}^{n-1})) \arrow[r] & 0 \arrow[r] & K_0(C^*(\mathbb{R}^n)) \arrow[d] \\ K_1(C^*(\mathbb{R}^n)) \arrow[u] & 0 \arrow[l] & K_1(C^*(\mathbb{R}^{n-1})), \arrow[l] \end{tikzcd}$$

from which the isomorphisms are immediate. The localised case is identical. $\square$

Example 5.2. Proposition 5.1 already shows that

$$K_0(C^*(\mathbb{R})) = 0, \qquad K_1(C^*(\mathbb{R})) \cong \mathbb{Z},$$

and then

$$K_0(C^*(\mathbb{R}^2)) \cong \mathbb{Z}, \qquad K_1(C^*(\mathbb{R}^2)) = 0.$$

In other words, each new Euclidean dimension swaps the roles of K_0 and K_1.

5.2 The computation

Theorem 5.3 (cf. [WY20, Thm. 7.5.1]). *For every $n \geq 0$ and $j \in \{0, 1\}$,*

$$K_j(C^*(\mathbb{R}^n)) \cong \begin{cases} \mathbb{Z}, & j \equiv n \pmod 2, \\ 0, & j \not\equiv n \pmod 2. \end{cases}$$

Equivalently,

$$K_0(C^*(\mathbb{R}^n)) \cong \begin{cases} \mathbb{Z}, & n \text{ even}, \\ 0, & n \text{ odd}, \end{cases} \qquad K_1(C^*(\mathbb{R}^n)) \cong \begin{cases} 0, & n \text{ even}, \\ \mathbb{Z}, & n \text{ odd}. \end{cases}$$

Proof. We argue by induction on n. If $n = 0$, then $\mathbb{R}^0$ is a point. By lemma 1.9, $C^*(\mathbb{R}^0) = \mathcal{K}$, and by [WY20, Example 2.7.3],

$$K_0(\mathcal{K}) \cong \mathbb{Z}, \qquad K_1(\mathcal{K}) = 0.$$

This is the base case. The induction step follows from proposition 5.1. $\square$

Corollary 5.4 ([WY20, Thm. 7.5.1]). *For every $n \geq 0$, the evaluation-at-one map*

$$\mathrm{ev}_* : K_*(C_L^*(\mathbb{R}^n)) \longrightarrow K_*(C^*(\mathbb{R}^n))$$

is an isomorphism.

Proof. The case $n = 0$ is the assembly map for a point, hence an isomorphism. For the inductive step, apply theorem 3.4 to the Roe and localised Roe covers $\mathbb{R}^n = E_n \cup F_n$. The outer terms vanish by proposition 4.1 and corollary 4.2, so the five lemma shows that if ev_* is an isomorphism for $\mathbb{R}^{n-1}$, then it is also an isomorphism for $\mathbb{R}^n$. $\square$

Definition 5.5 (cf. [WY20, Thm. 7.5.5])**.** Let $p \in \mathcal{K}$ be a rank-one projection, and set

$$\beta_0 := [p] \in K_0(C^*(\mathbb{R}^0)) = K_0(\mathcal{K}).$$

For $n \geq 1$, define β_n recursively by

$$\partial_n(\beta_n) = \beta_{n-1},$$

where ∂_n is the Mayer–Vietoris boundary isomorphism for

$$\mathbb{R}^n = \mathbb{R}^{n-1} \times (-\infty, 0] \cup \mathbb{R}^{n-1} \times [0, \infty).$$

The sign convention is the one we use below: the positive half-space is the I-term in the pushout boundary. Reversing the order of the two half-spaces reverses the sign. Thus $\beta_n \in K_n(C^*(\mathbb{R}^n))$, where $K_n(A) := K_{n \bmod 2}(A)$.

The recursive picture is

$$\beta_n \xrightarrow{\partial_n} \beta_{n-1} \xrightarrow{\partial_{n-1}} \cdots \xrightarrow{\partial_1} \beta_0 = [p].$$

The rest of these notes identify this generator more concretely.

6 The boundary map for $\mathbb{R}$ and the one-dimensional shift

6.1 How the boundary map is built

Suppose

$$\begin{tikzcd} I \cap J \arrow[r, hook, "\iota_I"] \arrow[d, hook, "\iota_J"'] & I \arrow[d, hook] \\ J \arrow[r, hook] & A \end{tikzcd}$$

is a pushout diagram of ideals with $A = I + J$. Then the inclusion $I \hookrightarrow A$ induces an isomorphism

$$I/(I \cap J) \cong A/J.$$

Hence there are short exact sequences

$$0 \longrightarrow I \cap J \longrightarrow I \longrightarrow I/(I \cap J) \longrightarrow 0$$

and

$$0 \longrightarrow J \longrightarrow A \longrightarrow A/J \longrightarrow 0.$$

The odd Mayer–Vietoris boundary is the composite

$$K_1(A) \longrightarrow K_1(A/J) \cong K_1(I/(I \cap J)) \xrightarrow{\partial} K_0(I \cap J),$$

where the last arrow is the index map for the first short exact sequence. The even boundary is defined in the same way, with K_0 and K_1 interchanged. Under the pushout formalism of [WY20, Prop. 2.7.15], these are the boundary maps in the six-term Mayer–Vietoris sequence.

6.2 The one-dimensional computation

Take

$$X = \mathbb{R}, \qquad E = (-\infty, 0], \qquad F = [0, \infty).$$

Then $E \cap F = \{0\}$, and the cover is coarsely excisive. By proposition 4.1,

$$K_*(C^*(E)) = K_*(C^*(F)) = 0.$$

Since $E \cap F$ is bounded, lemma 1.9 gives

$$C^*(E \cap F) = C^*(\{0\}) \cong \mathcal{K}.$$

Thus

$$K_0(C^*(E \cap F)) \cong \mathbb{Z}, \qquad K_1(C^*(E \cap F)) = 0.$$

The six-term sequence simplifies to

$$\begin{tikzcd}
K_0(\mathcal{K}) \arrow[r] & 0 \arrow[r] & K_0(C^*(\mathbb{R})) \arrow[d, "\partial_0"] \\
K_1(C^*(\mathbb{R})) \arrow[u, "\partial_1"] & 0 \arrow[l] & K_1(\mathcal{K}) = 0, \arrow[l]
\end{tikzcd}$$

so

$$K_0(C^*(\mathbb{R})) = 0, \qquad \partial_1 : K_1(C^*(\mathbb{R})) \xrightarrow{\cong} K_0(\mathcal{K})$$

is an isomorphism, and ∂_0 is the zero map.

Proposition 6.1 ([WY20, Example 2.7.3 and Thm. 7.5.5])**.** *Let $p \in \mathcal{K}$ be a rank-one projection. Then $[p]$ generates $K_0(\mathcal{K}) \cong \mathbb{Z}$. If $\beta \in K_1(C^*(\mathbb{R}))$ is the unique class satisfying*

$$\partial_1(\beta) = [p],$$

then β is a generator of $K_1(C^*(\mathbb{R})) \cong \mathbb{Z}$.

6.3 Toeplitz compression and the sign

We discretize $\mathbb{R}$ by $\mathbb{Z}$ to understand the generator. Let $S : \ell^2(\mathbb{Z}) \to \ell^2(\mathbb{Z})$ be the bilateral shift

$$S\delta_n = \delta_{n+1}.$$

On the non-ample module $\ell^2(\mathbb{Z})$, S is locally compact and has propagation one. On the ample module $\ell^2(\mathbb{Z}, H)$, where H is infinite-dimensional, the raw shift $S \otimes 1_H$ is not locally compact, because the matrix entries are 1_H. Therefore we use a rank-one projection $e \in \mathcal{K}(H)$ and the unitary representatives

$$U_{S^{-1}} := 1 + (S^{-1} - 1) \otimes e, \qquad U_S := 1 + (S - 1) \otimes e$$

in $C^*(\mathbb{Z})^+$. We write their K_1-classes as $[S^{-1}]$ and $[S]$. Through the coarse equivalence $\mathbb{Z} \hookrightarrow \mathbb{R}$, which induces an isomorphism on Roe algebra K-theory by [WY20, Thm. 5.1.15], we regard these as classes in $K_1(C^*(\mathbb{R}))$.

Let $H_+ := \ell^2(\mathbb{N})$, where $\mathbb{N} = \{0, 1, 2, \ldots\}$, and let

$$V : H_+ \hookrightarrow \ell^2(\mathbb{Z}), \qquad V\delta_n = \delta_n.$$

Let $P_+ := VV^*$ be the projection onto the positive half-line. From the Toeplitz discussion in [WY20, Sec. 3.4], if $f \in C(S^1)$ and C_f is the corresponding convolution operator, then

$$T_f := V^* C_f V$$

is the Toeplitz operator on H_+.

For $f(z) = z$, the convolution operator is S, so

$$T_z = V^* S V.$$

Thus T_z is the forward unilateral shift:

$$T_z \delta_n = \delta_{n+1}.$$

For $f(z) = z^{-1}$, the convolution operator is S^{-1}, so

$$T_{z^{-1}} = V^* S^{-1} V = T_z^*.$$

Thus T_z^* is the backward unilateral shift:

$$T_z^* \delta_0 = 0, \qquad T_z^* \delta_n = \delta_{n-1} \quad (n \geq 1).$$

Let p be the rank-one projection onto $\mathbb{C}\delta_0 \subset H_+$. Then

$$T_z^* T_z = 1, \qquad T_z T_z^* = 1 - p.$$

Therefore

$$\ker(T_z) = 0, \qquad \operatorname{coker}(T_z) \cong \mathbb{C}\delta_0,$$

and

$$\operatorname{Index}(T_z) = \dim \ker(T_z) - \dim \operatorname{coker}(T_z) = -1.$$

In $K_0(\mathcal{K})$ this says that the K-theory index class of T_z is $-[p]$. This agrees with the Toeplitz index theorem

$$\operatorname{Index}(T_f) = -\operatorname{wind}(f)$$

from [WY20, Thm. 3.4.1].

Similarly,

$$\ker(T_z^*) = \mathbb{C}\delta_0, \qquad \operatorname{coker}(T_z^*) = 0,$$

so

$$\operatorname{Index}(T_z^*) = 1,$$

and the K-theory index class of T_z^* is $[p]$.

Proposition 6.2 (cf. [WY20, Prop. 2.7.15, Example 2.8.3, and Thm. 3.4.1])**.** *With the boundary convention* *in definition 5.5 and proposition 6.1,*

$$\partial_1([S^{-1}]) = [p], \qquad \partial_1([S]) = -[p].$$

Hence

$$[S^{-1}] = \beta, \qquad [S] = -\beta.$$

Proof. This choice fixes the sign: the positive half-line F is the I-term and the negative half-line E is the J-term. In the pushout description, take

$$A = C^*(\mathbb{R}), \qquad I = C^*_X(F), \qquad J = C^*_X(E).$$

Modulo the negative half-line ideal J, the stabilised representative

$$U_{S^{-1}} = 1 + (S^{-1} - 1) \otimes e$$

compresses on the positive half-line to

$$T_z^* \otimes e + 1 \otimes (1 - e).$$

The second summand is invertible and contributes zero to the index. The first summand has index projection $p \otimes e$, which is identified with p under the standard stabilisation $\mathcal{K} \otimes \mathcal{K} \cong \mathcal{K}$. The odd Mayer–Vietoris boundary is the index map for

$$0 \longrightarrow I \cap J \longrightarrow I \longrightarrow I/(I \cap J) \longrightarrow 0.$$

Therefore

$$\partial_1([S^{-1}]) = \mathrm{Index}(T_z^* \otimes e + 1 \otimes (1 - e)) = [p].$$

The same argument for S gives the compression $T_z \otimes e + 1 \otimes (1 - e)$, whose index class is $-[p]$. □

7 Higher-dimensional shift products

7.1 The stabilised last-coordinate shift

Consider

$$\ell^2(\mathbb{Z}^n, H) \cong \ell^2(\mathbb{Z}^{n-1}) \otimes \ell^2(\mathbb{Z}) \otimes H \cong \ell^2(\mathbb{Z}^{n-1}, H) \otimes \ell^2(\mathbb{Z}),$$

where H is a separable infinite-dimensional Hilbert space, and the last factor records the last coordinate.

Let S_n be the bilateral shift in the last coordinate:

$$S_n \delta_{(m,k)} = \delta_{(m,k+1)}, \qquad S_n^{-1} \delta_{(m,k)} = \delta_{(m,k-1)},$$

where $m \in \mathbb{Z}^{n-1}$ and $k \in \mathbb{Z}$.

By [WY20, Thm. 5.1.15], the inclusion $\mathbb{Z}^n \hookrightarrow \mathbb{R}^n$ is a coarse equivalence and induces an isomorphism

$$K_*(C^*(\mathbb{Z}^n)) \cong K_*(C^*(\mathbb{R}^n)).$$

So we may work with $\mathbb{Z}^n$.

The Roe algebra $C^*(\mathbb{Z}^n)$ is the norm closure of $C[H_{\mathbb{Z}^n}]$, where $H_{\mathbb{Z}^n} := \ell^2(\mathbb{Z}^n, H)$ and $C[H_{\mathbb{Z}^n}]$ consists of bounded finite-propagation matrices

$$(T_{xy})_{x,y\in\mathbb{Z}^n}, \qquad T_{xy} \in \mathcal{K}(H).$$

The raw shift $S_n \otimes 1_H$ in the last coordinate has matrix entries

$$(S_n \otimes 1_H)_{(m,k),(m',k')} = \begin{cases} 1_H, & m = m',\ k = k' + 1, \\ 0, & \text{otherwise.} \end{cases}$$

Since 1_H is not compact, $S_n \otimes 1_H \notin C^*(\mathbb{Z}^n)$.

Let $e \in \mathcal{K}(H)$ be a rank-one projection. Then

$$((S_n^{-1} - 1) \otimes e)_{(m,k),(m',k')} = \begin{cases} -e, & (m,k) = (m',k'), \\ e, & m = m',\ k = k' - 1, \\ 0, & \text{otherwise.} \end{cases}$$

Now the matrix entries are compact and the propagation is finite. Hence

$$(S_n^{-1} - 1) \otimes e \in C^*(\mathbb{Z}^n),$$

and

$$U_{S_n^{-1}} := 1 + (S_n^{-1} - 1) \otimes e \in C^*(\mathbb{Z}^n)^+.$$

Equivalently,

$$U_{S_n^{-1}} = S_n^{-1} \otimes e + 1 \otimes (1 - e).$$

We have

$$U_{S_n^{-1}}^* = S_n \otimes e + 1 \otimes (1 - e).$$

Using

$$e(1-e) = 0, \qquad (1-e)e = 0, \qquad e^2 = e, \qquad (1-e)^2 = 1 - e,$$

we check that

$$U_{S_n^{-1}}^* U_{S_n^{-1}} = 1, \qquad U_{S_n^{-1}} U_{S_n^{-1}}^* = 1.$$

Thus $U_{S_n^{-1}}$ is a unitary in $C^*(\mathbb{Z}^n)^+$, and it defines

$$[U_{S_n^{-1}}] \in K_1(C^*(\mathbb{Z}^n)).$$

We write this class simply as

$$[S_n^{-1}] := [U_{S_n^{-1}}].$$

Similarly, set

$$U_{S_n} := 1 + (S_n - 1) \otimes e, \qquad [S_n] := [U_{S_n}].$$

From now on, when $S_n^{\pm 1}$ appears inside a Roe algebra K-theory formula, it denotes this stabilised unitary representative; the raw shift is only a mnemonic for the last-coordinate motion.

7.2 Defining the higher shift class by products

For $n \geq 2$, the product representative for the recursively defined class is

$$\beta_n = m_*(\beta_{n-1} \times [S_n^{-1}]).$$

This uses the stabilised last-coordinate representative from the previous subsection; the boundary computation below checks that this product representative has the recursive boundary value. Here the product is the external product on spectral K-theory

$$\times : \operatorname{sp} K_i(A) \otimes \operatorname{sp} K_j(B) \longrightarrow \operatorname{sp} K_{i+j}(A \widehat{\otimes}_{\max} B)$$

from [WY20, Def. 2.10.7]. We take

$$A = C^*(\mathbb{Z}^{n-1}), \qquad B = C^*(\mathbb{Z}).$$

By [WY20, Prop. 2.9.12], spectral K-theory agrees with ordinary K-theory for the inner-graded algebras used here.

The product decomposition

$$\mathbb{Z}^n = \mathbb{Z}^{n-1} \times \mathbb{Z}$$

gives commuting multiplier representations of the two Roe algebras on the product module. By the universal property of the maximal tensor product, these induce a product-coordinate $*$-homomorphism

$$m : C^*(\mathbb{Z}^{n-1}) \widehat{\otimes}_{\max} C^*(\mathbb{Z}) \longrightarrow C^*(\mathbb{Z}^n).$$

On finite-propagation elementary tensors, this is the tensor-product operator on the product module, after the standard stabilisation

$$\mathcal{K}(H) \otimes \mathcal{K}(H) \cong \mathcal{K}(H).$$

Propagation remains finite, and compact matrix entries tensor to compact matrix entries. The separate factors $a \otimes 1$ and $1 \otimes b$ are generally multipliers rather than Roe operators, but elementary products $a \otimes b$ are Roe operators on the product module; this is the map that extends to m. Thus the external product first gives a class in the tensor product, and m_* gives a class in the Roe algebra of $\mathbb{Z}^n$. In this notation, $[S_n^{-1}]$ is the one-dimensional shift class in the last coordinate; after applying m_*, it is represented by the last-coordinate shift on $\mathbb{Z}^n$.

If n is odd, then $\beta_{n-1} \in K_0(C^*(\mathbb{Z}^{n-1}))$. For a projection q representing a K_0-class, the product

$$[q] \times [S_n^{-1}]$$

is represented by

$$U_n^q := q \otimes U_{S_n^{-1}} + (1 - q) \otimes 1,$$

where $U_{S_n^{-1}} = 1 + (S_n^{-1} - 1) \otimes e$ is the stabilised shift representative. Indeed,

$$(U_n^q)^* = q \otimes U_{S_n} + (1 - q) \otimes 1,$$

and

$$U_n^q (U_n^q)^* = (U_n^q)^* U_n^q = 1.$$

If

$$\chi = [q] - [q_0] \in K_0(C^*(\mathbb{Z}^{n-1})),$$

then

$$\chi \times [S_n^{-1}]$$

is represented by

$$U_n^q (U_n^{q_0})^{-1}.$$

If n is even, then $\beta_{n-1} \in K_1(C^*(\mathbb{Z}^{n-1}))$. Let u be a unitary representative. The product

$$[u] \times [S_n^{-1}] \in K_0(C^*(\mathbb{Z}^n))$$

is produced as a projection using the spectral picture. Namely, let

$$\phi_u : C_0(\mathbb{R}) \longrightarrow C^*(\mathbb{Z}^{n-1}) \widehat{\otimes} \operatorname{Cliff}_{\mathbb{C}}(\mathbb{R}) \widehat{\otimes} \mathcal{K}$$

and

$$\phi_{S_n^{-1}} : C_0(\mathbb{R}) \longrightarrow C^*(\mathbb{Z}) \widehat{\otimes} \operatorname{Cliff}_{\mathbb{C}}(\mathbb{R}) \widehat{\otimes} \mathcal{K}$$

represent the two K_1-classes. The external product is defined using the comultiplication

$$\Delta : C_0(\mathbb{R}) \longrightarrow C_0(\mathbb{R}) \widehat{\otimes} C_0(\mathbb{R}).$$

Then

$$\phi := (\phi_u \widehat{\otimes} \phi_{S_n^{-1}}) \circ \Delta$$

represents a class in

$$\operatorname{sp} K_2(C^*(\mathbb{Z}^{n-1}) \widehat{\otimes}_{\max} C^*(\mathbb{Z})).$$

By Bott periodicity in the spectral picture, this is the same parity as $\operatorname{sp} K_0$. Since the grading is inner, [WY20, Prop. 2.9.12] identifies this with ordinary K_0.

Let u_ϕ be the Cayley transform of ϕ, and let

$$v = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

be the grading multiplier. Set

$$p_\phi := \frac{1}{2}(v u_\phi + 1).$$

Then

$$p_\phi \equiv \frac{1}{2}(v + 1) \pmod{(C^*(\mathbb{Z}^{n-1}) \widehat{\otimes}_{\max} C^*(\mathbb{Z})) \widehat{\otimes} \mathcal{K}}.$$

Thus the corresponding ordinary K_0-class is

$$\left[\frac{1}{2}(v+1)\right] - [p_\phi] \in K_0(C^*(\mathbb{Z}^{n-1})\widehat{\otimes}_{\max} C^*(\mathbb{Z})).$$

After applying m_*, this gives a class in $K_0(C^*(\mathbb{Z}^n))$. For example,

$$\beta_2 = m_*([S_1^{-1}] \times [S_2^{-1}]) = m_*\left(\left[\frac{1}{2}(v+1)\right] - [p_\phi]\right) \in K_0(C^*(\mathbb{Z}^2)) \cong K_0(C^*(\mathbb{R}^2)).$$

7.3 Boundary computation for the odd-dimensional step

We give the concrete computation for the $K_0 \times K_1 \to K_1$ case. Work on

$$\ell^2(\mathbb{Z}^n, H) \cong \ell^2(\mathbb{Z}^{n-1}) \otimes \ell^2(\mathbb{Z}) \otimes H.$$

Let

$$\chi = [q] - [q_0] \in K_0(C^*(\mathbb{Z}^{n-1})),$$

where q, q_0 are projections after stabilising if necessary. In the non-unital case, they are understood in the unitisation, with matching scalar parts.

Define

$$U_n^q := q \otimes U_{S_n^{-1}} + (1-q) \otimes 1, \qquad U_n^{q_0} := q_0 \otimes U_{S_n^{-1}} + (1-q_0) \otimes 1.$$

Then the class $\chi \times [S_n^{-1}]$ is represented by

$$U_n^q (U_n^{q_0})^{-1}.$$

Let

$$V : \ell^2(\mathbb{N}) \hookrightarrow \ell^2(\mathbb{Z}), \qquad V\delta_k = \delta_k,$$

and put

$$P_+ := VV^*, \qquad P_n^+ := 1 \otimes P_+ \otimes 1_H.$$

Identifying the range of P_n^+ with

$$\ell^2(\mathbb{Z}^{n-1}) \otimes \ell^2(\mathbb{N}) \otimes H,$$

the positive-half-space compression of U_n^q is

$$\widehat{U} = q \otimes \big(T_z^* \otimes e + 1 \otimes (1-e)\big) + (1-q) \otimes 1.$$

The summand with $1-e$ is invertible and has zero index contribution. Suppressing this harmless stabilising summand, the index computation is the same as for

$$\widehat{U} = q \otimes T_z^* + (1-q) \otimes 1.$$

Take

$$W := q \otimes T_z + (1-q) \otimes 1$$

as a parametrix for this reduced operator. Since $T_z^* T_z = 1$ and $T_z T_z^* = 1 - p$, where p is the rank-one projection onto $\mathbb{C}\delta_0$, we get

$$\widehat{U}W = 1, \qquad W\widehat{U} = 1 - q \otimes p.$$

Thus

$$1 - \widehat{U}W = 0, \qquad 1 - W\widehat{U} = q \otimes p.$$

Using the algebraic index formula from [WY20, Rem. 2.8.2, Eq. (2.20)],

$$\operatorname{Ind}[\widehat{U}] = \left[\begin{pmatrix} (1 - W\widehat{U})^2 & W(1 - \widehat{U}W) \\ \widehat{U}(2 - W\widehat{U})(1 - W\widehat{U}) & \widehat{U}W(2 - \widehat{U}W) \end{pmatrix}\right] - \left[\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\right].$$

Substituting

$$1 - W\widehat{U} = q \otimes p, \qquad 1 - \widehat{U}W = 0, \qquad \widehat{U}W = 1,$$

we obtain

$$(1 - W\widehat{U})^2 = q \otimes p, \qquad W(1 - \widehat{U}W) = 0,$$

and

$$\widehat{U}W(2 - \widehat{U}W) = 1.$$

For the lower-left entry,

$$(2 - W\widehat{U})(1 - W\widehat{U}) = (1 + q \otimes p)(q \otimes p) = 2q \otimes p.$$

Hence

$$\widehat{U}(2 - W\widehat{U})(1 - W\widehat{U}) = 2\widehat{U}(q \otimes p) = 2q \otimes T_z^* p = 0,$$

because $T_z^*\delta_0 = 0$. Therefore

$$\operatorname{Ind}[\widehat{U}] = \left[\begin{pmatrix} q \otimes p & 0 \\ 0 & 1 \end{pmatrix}\right] - \left[\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\right] = [q \otimes p].$$

Similarly,

$$\operatorname{Ind}[\widehat{U}^{q_0}] = [q_0 \otimes p].$$

So

$$\operatorname{Ind}[\widehat{U}^{q}(\widehat{U}^{q_0})^{-1}] = [q \otimes p] - [q_0 \otimes p].$$

By stabilisation, this corresponds to

$$[q] - [q_0] = \chi.$$

Therefore

$$\partial_n(m_*(\chi \times [S_n^{-1}])) = \chi.$$

Taking $\chi = \beta_{n-1}$ gives

$$\partial_n(\beta_n) = \beta_{n-1}, \qquad \beta_n = m_*(\beta_{n-1} \times [S_n^{-1}])$$

in the odd-dimensional step where $\beta_{n-1} \in K_0$.

For the forward shift S_n, the compression is

$$\widehat{U}_{\mathrm{forw}} = q \otimes T_z + (1-q) \otimes 1,$$

and a parametrix is

$$W_{\mathrm{forw}} = q \otimes T_z^* + (1-q) \otimes 1.$$

Then

$$W_{\mathrm{forw}}\widehat{U}_{\mathrm{forw}} = 1, \qquad \widehat{U}_{\mathrm{forw}}W_{\mathrm{forw}} = 1 - q \otimes p.$$

Thus

$$1 - W_{\mathrm{forw}}\widehat{U}_{\mathrm{forw}} = 0, \qquad 1 - \widehat{U}_{\mathrm{forw}}W_{\mathrm{forw}} = q \otimes p.$$

Using the same algebraic index formula,

$$(1 - W_{\mathrm{forw}}\widehat{U}_{\mathrm{forw}})^2 = 0,$$

$$W_{\mathrm{forw}}(1 - \widehat{U}_{\mathrm{forw}}W_{\mathrm{forw}}) = W_{\mathrm{forw}}(q \otimes p) = q \otimes T_z^* p = 0,$$

$$\widehat{U}_{\mathrm{forw}}(2 - W_{\mathrm{forw}}\widehat{U}_{\mathrm{forw}})(1 - W_{\mathrm{forw}}\widehat{U}_{\mathrm{forw}}) = 0,$$

and

$$\widehat{U}_{\mathrm{forw}}W_{\mathrm{forw}}(2 - \widehat{U}_{\mathrm{forw}}W_{\mathrm{forw}}) = (1 - q \otimes p)(1 + q \otimes p) = 1 - q \otimes p.$$

Therefore the index projection becomes

$$\left[\begin{pmatrix} 0 & 0 \\ 0 & 1 - q \otimes p \end{pmatrix}\right] - \left[\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\right] = -[q \otimes p].$$

Therefore

$$\partial_n(m_*(\chi \times [S_n])) = -\chi.$$

So the backward shift gives the positive boundary value, and the forward shift gives the negative boundary value.

For the even-dimensional step, the same compatibility with the boundary map follows by suspending this calculation in the spectral picture. The explicit matrix computation above is the $K_0 \times K_1 \to K_1$ case; the $K_1 \times K_1 \to K_0$ case is represented by the corresponding projection model, and the conclusion is transported by naturality of the external product and Bott periodicity [WY20, Defs. 2.10.4 and 2.10.7, Prop. 2.9.12].

8 The Dirac picture and comparison with the shift generator

8.1 The Euclidean Dirac class

Definition 8.1 ([WY20, Def. 9.3.1])**.** Let $n \in \mathbb{N}$, and let H_n be the Hilbert space underlying the Clifford algebra $\mathrm{Cliff}_{\mathbb{C}}(\mathbb{R}^n)$. Fix an orthonormal basis $\{e_1, \ldots, e_n\}$ for $\mathbb{R}^n$, and let $x_1, \ldots, x_n$ be the corresponding coordinates. For $v \in \mathbb{R}^n$, write $\widehat{v}$ for the operator on $\mathrm{Cliff}_{\mathbb{C}}(\mathbb{R}^n)$ defined on a homogeneous Clifford element w by

$$\widehat{v}(w) = (-1)^{\deg(w)} wv.$$

Thus $\widehat{v}$ is right multiplication by v, twisted by the grading. The Dirac operator D_n on $L^2(\mathbb{R}^n; H_n)$ has domain $\mathcal{S}(\mathbb{R}^n; H_n)$ and is given by

$$(D_n u)(x) = \sum_{i=1}^{n} \widehat{e_i} \frac{\partial u}{\partial x_i}(x).$$

The Bott operator C_n is

$$(C_n u)(x) = \sum_{i=1}^{n} e_i x_i u(x).$$

Let $\psi : \mathbb{R} \to [-1, 1]$ be a continuous odd function with

$$\lim_{x\to\pm\infty} \psi(x) = \pm 1.$$

Set

$$\mathcal{F}_n := \psi(D_n).$$

By the functional calculus and the index construction used for Dirac operators, $\mathcal{F}_n$ is a self-adjoint multiplier and $1 - \mathcal{F}_n^2 \in C^*(\mathbb{R}^n)$; hence $\mathcal{F}_n$ defines an index class [WY20, Lem. 3.2.7 and Def. 2.8.5]. In even dimensions,

$$\mathrm{Index}[\mathcal{F}_{2m}] \in K_0(C^*(\mathbb{R}^{2m})),$$

and in odd dimensions,

$$\mathrm{Index}[\mathcal{F}_{2m+1}] \in K_1(C^*(\mathbb{R}^{2m+1})).$$

Let

$$[D_n] \in K_n(L^*(\mathbb{R}^n)) = K_n(\mathbb{R}^n)$$

be the K-homology Dirac class, and let

$$[C_n] \in K^n(\mathbb{R}^n)$$

be the Bott class. By [WY20, Lem. 9.3.4],

$$[D_n] = [D_1] \times \cdots \times [D_1], \qquad [C_n] = [C_1] \times \cdots \times [C_1],$$

where the products are exterior products. By [WY20, Thm. 9.3.5],

$$\langle [D_n], [C_n] \rangle = 1 \in \mathbb{Z}.$$

If $\alpha \in K_n(\mathbb{R}^n)$ and $\gamma \in K^n(\mathbb{R}^n)$ are generators, and if

$$[D_n] = m\alpha, \qquad [C_n] = r\gamma,$$

then

$$1 = \langle [D_n], [C_n] \rangle = mr\langle \alpha, \gamma \rangle.$$

Hence $|m| = |r| = 1$, and $[D_n]$ is a generator of $K_n(\mathbb{R}^n)$.

By corollary 5.4, the evaluation-at-one map

$$\mathrm{ev} : C_L^*(\mathbb{R}^n) \longrightarrow C^*(\mathbb{R}^n)$$

induces an isomorphism on K-theory. Also, by proposition 1.8, the localised Roe algebra computes K-homology for ample modules. Therefore the Dirac class gives a localised index class represented by the family

$$\mathcal{F}^L_{n,t} := \psi(t^{-1}D_n), \qquad t \in [1,\infty),$$

namely

$$\mathrm{Index}_L[(\mathcal{F}^L_{n,t})] \in K_n(C^*_L(\mathbb{R}^n)),$$

and evaluation at $t = 1$ gives

$$\mathrm{ev}_*(\mathrm{Index}_L[(\mathcal{F}^L_{n,t})]) = \mathrm{Index}[\mathcal{F}_n] \in K_n(C^*(\mathbb{R}^n)).$$

Since $[D_n]$ is a generator and ev_* is an isomorphism, $\mathrm{Index}[\mathcal{F}_n]$ is a generator. With the one-dimensional sign convention fixed above, we choose the positive generator so that

$$\beta_n = \mathrm{Index}[\mathcal{F}_n] = \mathrm{Index}[\psi(D_n)].$$

Under the decomposition $\mathbb{R}^n = \mathbb{R}^{n-1} \times \mathbb{R}$, [WY20, Rem. 9.3.2] gives

$$D_n = D_{n-1}\widehat{\otimes}1 + 1\widehat{\otimes}D_1.$$

This is compatible with

$$\partial_n(\mathrm{Index}[\mathcal{F}_n]) = \mathrm{Index}[\mathcal{F}_{n-1}], \qquad \partial_n(\beta_n) = \beta_{n-1}.$$

8.2 Comparison with the shift product

The dimension-one calculation fixed the sign convention:

$$[S^{-1}] \longleftrightarrow \mathrm{Index}[\psi(D_1)].$$

Inductively, the product construction gives

$$m_*([S_1^{-1}] \times \cdots \times [S_n^{-1}]) \longleftrightarrow \mathrm{Index}[\psi(D_n)].$$

This is compatible with the Dirac product formula above; more precisely, for the last-coordinate splitting, $D_n = D_{n-1}\widehat{\otimes}1 + 1\widehat{\otimes}D_1$. Thus the shift product picture and the Dirac picture give the same generator, with the sign convention fixed by

$$\partial_n(\beta_n) = \beta_{n-1}.$$

In particular, the positive generator is represented by the backward shift in the last coordinate.

8.3 Why the Dirac operator looks like a shift

The harmonic oscillator is

$$H = -\frac{d^2}{dx^2} + x^2 - 1$$

on $L^2(\mathbb{R})$, with domain the Schwartz-class functions $\mathcal{S}(\mathbb{R})$ [WY20, Def. D.3.1]. The annihilation and creation operators are

$$A = x + \frac{d}{dx}, \qquad A^* = x - \frac{d}{dx},$$

considered on the domain $\mathcal{S}(\mathbb{R})$, and they satisfy

$$H = A^*A = AA^* - 2.$$

Define

$$\psi_0(x) = \pi^{-1/4}e^{-x^2/2}, \qquad \psi_k = \frac{1}{\sqrt{2^k k!}}(A^*)^k\psi_0.$$

By [WY20, Prop. D.3.3], $(\psi_k)_{k=0}^\infty$ is an orthonormal basis of $L^2(\mathbb{R})$ consisting of eigenvectors for H, with

$$H\psi_k = 2k\psi_k.$$

We compute

$$\psi_0'(x) = -x\psi_0(x),$$

so

$$A\psi_0 = \left(\frac{d}{dx} + x\right)\psi_0 = \psi_0' + x\psi_0 = 0.$$

For $n \geq 1$,

$$A\psi_n = \sqrt{2n}\,\psi_{n-1}.$$

If $\psi_{-1} := 0$, then this formula holds for all $n \geq 0$. Thus

$$A\psi_0 = 0, \qquad A\psi_1 = \sqrt{2}\,\psi_0, \qquad A\psi_2 = 2\psi_1, \qquad A\psi_3 = \sqrt{6}\,\psi_2, \ldots$$

Compare this with the backward unilateral shift T_z^*:

$$T_z^*\delta_0 = 0, \qquad T_z^*\delta_1 = \delta_0, \qquad T_z^*\delta_2 = \delta_1, \ldots$$

Thus, in the Hermite basis, A is a weighted backward shift. The one-dimensional Bott–Dirac operator appearing in the proof of Bott periodicity is

$$B = \begin{pmatrix} 0 & A^* \\ A & 0 \end{pmatrix} = \begin{pmatrix} 0 & -\frac{d}{dx} + x \\ \frac{d}{dx} + x & 0 \end{pmatrix},$$

as in [WY20, proof of Thm. 9.3.5]. Since $A\psi_0 = 0$ and $A\psi_n = \sqrt{2n}\psi_{n-1}$ for $n \geq 1$,

$$\ker(A) = \mathbb{C}\psi_0.$$

Moreover, every basis vector ψ_k is hit by A because

$$A\left(\frac{1}{\sqrt{2(k+1)}}\psi_{k+1}\right) = \psi_k.$$

So in the Schwartz-class model, $A : \mathcal{S}(\mathbb{R}) \to \mathcal{S}(\mathbb{R})$ is surjective with one-dimensional kernel, and

$$\operatorname{Index}(A) = \dim\ker(A) - \dim\operatorname{coker}(A) = 1.$$

This is the same index pattern as the backward unilateral shift:

$$\operatorname{Index}(T_z^*) = 1.$$

So we have the correspondences

$$T_z^* \longleftrightarrow A = \frac{d}{dx} + x, \qquad \delta_0 \longleftrightarrow \psi_0(x) = \pi^{-1/4} e^{-x^2/2}, \qquad [p_{\delta_0}] \longleftrightarrow [p_{\psi_0}].$$

Thus the Toeplitz generator and the Bott–Dirac generator have the same index-theoretic pattern.

9 Assembly, Rips complexes, and coarse K-homology

This section records the background used later, without repeating the definitions of Roe and localised Roe algebras from definitions 1.4 and 1.7.

9.1 Assembly maps and coarse K-homology

Let X be a proper metric space, and assume first that the group action is trivial. The evaluation-at-one $*$-homomorphism

$$\mathrm{ev} : C_L^*(H_X) \longrightarrow C^*(H_X), \qquad (T_t) \longmapsto T_1$$

induces, after the identification of proposition 1.8, the assembly map.

Definition 9.1 ([WY20, Def. 7.1.1])**.** The *assembly map*, or *higher index map*, for X is

$$\mu_X : K_*(X) \longrightarrow K_*(C^*(X)).$$

It is induced by the evaluation-at-one map on localised Roe algebras.

In the equivariant setting, X is equipped with a proper isometric action of a countable discrete group G, and the same definitions use equivariant modules and equivariant Roe algebras. We then write

$$\mu_X : K_*^G(X) \longrightarrow K_*(C^*(X)^G).$$

Definition 9.2 ([WY20, Def. 7.1.4])**.** The equivariant bounded category over X, denoted $\mathcal{C}^G(X)$, has objects pairs (Y, p), where Y is a proper metric space with a proper isometric G-action and $p : X \to Y$ is an equivariant coarse equivalence. A morphism from (Y, p_Y) to (Z, p_Z) is a continuous equivariant coarse map $f : Y \to Z$ such that $f \circ p_Y$ is close to p_Z.

Definition 9.3 ([WY20, Defs. 7.1.7, 7.1.8, and 7.1.10])**.** The equivariant coarse K-homology of X, denoted

$$KX_*^G(X),$$

is the universal F-group for the functor

$$F(Y) := K_*^G(Y)$$

on $C^G(X)$. For an object (Y, p) in $C^G(X)$, the X-assembly map is

$$\mu_{Y,X} : K_*^G(Y) \xrightarrow{\mu_Y} K_*(C^*(Y)^G) \xrightarrow{(p_*)^{-1}} K_*(C^*(X)^G).$$

The Baum–Connes assembly map is the homomorphism

$$\mu : KX_*^G(X) \longrightarrow K_*(C^*(X)^G)$$

induced by the universal property of $KX_*^G(X)$. The Baum–Connes conjecture for X says that this map is an isomorphism. If $G = \{e\}$, we omit G and call

$$\mu : KX_*(X) \longrightarrow K_*(C^*(X))$$

the coarse Baum–Connes assembly map.

9.2 Rips complexes and the direct-limit model

Definition 9.4 ([WY20, Def. A.3.10])**.** Let $r \in (0, \infty)$. A subset $Z \subseteq X$ is r-separated if

$$d(x, y) \geq r \qquad \text{for all distinct } x, y \in Z.$$

It is an r-net if it is r-separated and, for every $x \in X$, there is $z \in Z$ with

$$d(x, z) < r.$$

By [WY20, Lem. A.3.11], every proper metric space admits an r-net for any $r \in (0, \infty)$. Moreover, the restricted metric on any net is proper, and the inclusion

$$Z \hookrightarrow X$$

is a coarse equivalence.

Definition 9.5 ([WY20, Defs. 7.2.1, 7.2.2, and 7.2.8])**.** If F is a finite set, let $\sigma(F)$ be the set of formal sums

$$\sum_{z \in F} t_z z, \qquad t_z \in [0, 1], \qquad \sum_{z \in F} t_z = 1.$$

Let $S(\mathbb{R}^F)$ be the unit sphere in the finite-dimensional Euclidean space $\mathbb{R}^F$ spanned by F. Embed $\sigma(F)$ into $S(\mathbb{R}^F)$ by

$$\sum_{z \in F} t_z z \longmapsto \left(\sum_{z \in F} t_z^2 \right)^{-1/2} \sum_{z \in F} t_z z.$$

The spherical metric on $\sigma(F)$ is

$$d(x, y) := \frac{2}{\pi} d_{\text{in}}(x, y),$$

where d_{in} is the intrinsic metric on the sphere after the above embedding. The factor $2/\pi$ normalises the diameter of each spherical simplex to be 1.

Let Z be a locally finite metric space and let $r \geq 0$. The spherical Rips complex $S_r(Z)$ consists of all formal sums

$$x = \sum_{z \in Z} t_z z$$

such that $t_z \in [0,1]$, $\sum_z t_z = 1$, and $\operatorname{diam}(\operatorname{supp}(x)) \leq r$, where $\operatorname{supp}(x) := \{z \in Z : t_z \neq 0\}$. If $F \subseteq Z$ is finite and $\operatorname{diam}(F) \leq r$, the simplex spanned by F is the subset $\sigma(F) \subseteq S_r(Z)$ of formal sums supported in F, with the spherical metric.

A simplicial path γ from x to y in $S_r(Z)$ is a finite sequence $x = x_0, x_1, \ldots, x_n = y$, together with simplices $\sigma_1, \ldots, \sigma_n$ such that σ_i contains x_{i-1} and x_i. Its length is

$$\ell(\gamma) := \sum_{i=1}^{n} d_{\sigma_i}(x_{i-1}, x_i),$$

and the spherical distance is

$$d_{S_r}(x,y) := \inf\{\ell(\gamma) : \gamma \text{ is a simplicial path from } x \text{ to } y\},$$

with $d_{S_r}(x,y) = \infty$ if no simplicial path exists.

A semi-simplicial path δ from x to y in $S_r(Z)$ is a sequence

$$x = x_0, y_0, x_1, y_1, \ldots, x_n, y_n = y,$$

where $x_1, \ldots, x_n \in Z$ and $y_0, \ldots, y_{n-1} \in Z$. Its length is

$$\ell(\delta) := \sum_{i=0}^{n} d_{S_r}(x_i, y_i) + \sum_{i=0}^{n-1} d_Z(y_i, x_{i+1}).$$

The semi-spherical distance is

$$d_{P_r}(x,y) := \inf\{\ell(\delta) : \delta \text{ is a semi-simplicial path from } x \text{ to } y\}.$$

The Rips complex $P_r(Z)$ is the set $S_r(Z)$ equipped with the metric d_{P_r}. In particular, $P_0(Z) = Z$.

Proposition 9.6 ([WY20, Prop. 7.2.11])**.** *If Z is countable and locally finite, then $P_r(Z)$ is a proper, second countable metric space. For every $s \geq r$, the canonical inclusion*

$$i_{rs} : P_r(Z) \longrightarrow P_s(Z)$$

is a homeomorphism onto its image and a coarse equivalence.

Theorem 9.7 ([WY20, Thm. 7.2.16])**.** *Let X be a proper metric space with an isometric action of a countable group G, and let $Z \subseteq X$ be a locally finite G-invariant net. Then*

$$KX_*^G(X) \cong \lim_{r\to\infty} K_*^G(P_r(Z)).$$

Under this identification, the Baum–Connes assembly map is the direct limit of the X-assembly maps

$$\lim_{r\to\infty} \mu_{P_r(Z),X} : \lim_{r\to\infty} K_*^G(P_r(Z)) \longrightarrow K_*(C^*(X)^G).$$

If $G = \{e\}$, then

$$KX_*(X) \cong \lim_{r\to\infty} K_*(P_r(Z))$$

and

$$\mu = \lim_{r\to\infty} \mu_{P_r(Z),X} : \lim_{r\to\infty} K_*(P_r(Z)) \longrightarrow K_*(C^*(X)).$$

9.3 Generators from Rips complexes

When the coarse assembly map for X is an isomorphism, every class

$$\alpha \in K_i(C^*(X))$$

has the form

$$\alpha = \mu_{P_r(Z),X}(\alpha_r)$$

for some scale $r \geq 0$ and some

$$\alpha_r \in K_i(P_r(Z)).$$

Under this identification, representatives

$$\alpha_r \in K_i(P_r(Z)), \qquad \beta_s \in K_i(P_s(Z))$$

represent the same element of $K_i(C^*(X))$ if and only if there is $t \geq \max\{r, s\}$ such that

$$(i_{rt})_*(\alpha_r) = (i_{st})_*(\beta_s) \qquad \text{in } K_i(P_t(Z)).$$

This is exactly the equivalence relation in the direct limit. Thus $K_i(P_r(Z))$ may have several independent classes at a fixed scale, but the maps

$$(i_{rs})_* : K_i(P_r(Z)) \longrightarrow K_i(P_s(Z))$$

may identify or kill some of these classes. The resulting direct limit may be noncyclic.

9.4 Bounded geometry and coarse assembly

Definition 9.8 ([WY20, Def. A.3.19])**.** A metric space X has bounded geometry if, for every $r \in (0, \infty)$, there is $n_r \in \mathbb{N}$ such that

$$|B(x; r)| \leq n_r \qquad \text{for all } x \in X.$$

In particular, a bounded geometry metric space is proper and discrete.

Theorem 9.9 ([WY20, Thm. 12.0.2])**.** *If X is a bounded geometry metric space that coarsely embeds into a Hilbert space, then the coarse assembly map*

$$\mu : KX_*(X) \xrightarrow{\cong} K_*(C^*(X))$$

is an isomorphism. Therefore, for $i = 0, 1$,

$$K_i(C^*(X)) \cong KX_i(X) \cong \lim_{r\to\infty} K_i(P_r(Z)),$$

where $Z \subseteq X$ is any locally finite net.

Definition 9.10 ([WY20, Defs. 7.3.1 and 7.3.4])**.** A proper metric space X is *uniformly contractible* if, for every $r \geq 0$, there is $s \geq r$ such that, for every $x \in X$, the inclusion

$$B(x;r) \hookrightarrow B(x;s)$$

is homotopic to a constant map. A simplicial complex structure on X is *good* if it is finite-dimensional, if the vertex set has bounded geometry, and if the inclusion of the vertex set into X is a coarse equivalence.

Theorem 9.11 ([WY20, Thm. 7.3.6])**.** *If X is a uniformly contractible good simplicial complex, then the coarse Baum–Connes conjecture for X identifies with the ordinary assembly map*

$$\mu_X : K_*(X) \longrightarrow K_*(C^*(X)).$$

So if X is uniformly contractible and good, and if the coarse assembly map is known to be an isomorphism, then

$$K_i(C^*(X)) \cong K_i(X), \qquad i = 0, 1.$$

9.5 Elliptic operators and K-homology classes

We now recall how elliptic differential operators give analytic K-homology classes.

Definition 9.12 ([WY20, Def. 8.3.1])**.** Let D be a differential operator on a Hermitian bundle $S \to M$. The operator D is elliptic if its symbol $\sigma_D(x, \xi) : S_x \to S_x$ is invertible for every nonzero cotangent vector $\xi \in T_x^*M$.

A differentiable bounded function $f : \mathbb{R} \to \mathbb{C}$ has slow oscillation at infinity if $\sup_{x\in\mathbb{R}} |xf'(x)| < \infty$. The algebra $C_{\mathrm{so}}(\mathbb{R})$ is the C^*-subalgebra of $C_b(\mathbb{R})$ generated by bounded functions with slow oscillation at infinity [WY20, Def. 8.2.4].

By [WY20, Lem. 8.2.8], there is a family $(g_t)_{t\in[1,\infty)}$ of smooth functions

$$g_t : M \longrightarrow [0, 1]$$

such that:

(i) the map $t \mapsto g_t$ is norm continuous as a function $[1, \infty) \to C_b(M)$;

(ii) for every compact $K \subseteq M$, there is t_K such that $g_t = 1$ on K for all $t \geq t_K$;

(iii) each operator $g_t D g_t$, with domain $C_c^\infty(M; S)$, is essentially self-adjoint.

If $f \in C_{\mathrm{so}}(\mathbb{R})$, then the pair $(f, (g_t))$ is multiplier data for D in the sense of [WY20, Def. 8.2.9].

Theorem 9.13 ([WY20, Thm. 8.2.10 and Thm. 8.3.3])**.** *Let D be a formally self-adjoint differential operator, and let $(f, (g_t))$ be multiplier data for D. Following the scaled multiplier construction, define*

$$F_t := f(t^{-1} g_t D g_t).$$

Then $(F_t)_{t\in[1,\infty)}$ defines a multiplier of $L^*(L^2(M;S))$. If D is elliptic and, in addition, $f \in C_0(\mathbb{R})$, then

$$(F_t)_{t\in[1,\infty)} \in L^*(L^2(M;S)).$$

Proposition 9.14 ([WY20, Thm. 8.2.10, Thm. 8.3.3, Constructions 8.3.11 and 8.3.12, Prop. 8.3.13])**.** *Let D be a formally self-adjoint elliptic differential operator as in definition 9.12, and choose a family (g_t) as above. In the odd case, the scaled functional-calculus assignment*

$$\Phi : C_0(\mathbb{R}) \longrightarrow L^*(L^2(M;S)), \qquad h \longmapsto \big(h(t^{-1}g_t D g_t)\big)_{t\in[1,\infty)}$$

is a $*$-homomorphism by [WY20, Construction 8.3.11], using theorem 9.13 together with functional calculus. Tensoring with a rank-one projection $p \in \mathcal{K}$ gives

$$\Phi \otimes p : C_0(\mathbb{R}) \longrightarrow L^*(L^2(M;S)) \otimes \mathcal{K},$$

which defines

$$[D] \in K_1(L^*(L^2(M;S))).$$

If $L^2(M;S)$ is an ample M module, then this is a class

$$[D] \in K_1(M).$$

In the even case, assume

$$S = S^- \oplus S^+$$

is $\mathbb{Z}/2$-graded, and that D interchanges S^- and S^+. Let U be the grading operator on

$$L^2(M;S) = L^2(M;S^-) \oplus L^2(M;S^+),$$

acting by multiplication by -1 on $L^2(M;S^-)$ and by $+1$ on $L^2(M;S^+)$. Conjugation by U gives an inner grading on

$$L^*(L^2(M;S)),$$

and D is odd for this grading. The algebra $C_0(\mathbb{R})$ is graded by even and odd functions. Tensoring the graded $*$-homomorphism above with an even rank-one projection in the graded compact operators gives a class

$$[D] \in K_0(L^*(L^2(M;S))).$$

If $L^2(M;S)$ is an ample M module, this identifies with a class in $K_0(M)$. The resulting classes $[D] \in K_i(M)$ do not depend on the choices of the family (g_t) or the rank-one projection. In the index-theoretic representative of [WY20, Rem. 8.3.15], the normalising function may also be varied through admissible choices.

9.6 Boundary maps at the level of coarse K-homology

Let

$$X = E \cup F$$

be a coarsely excisive decomposition by closed subsets. Exercise 7.6.14 of [WY20] gives a commutative diagram of Mayer–Vietoris sequences:

$$\begin{tikzcd}
KX_i(E\cap F) \arrow[r] \arrow[d, "\mu_{E\cap F}"'] & KX_i(E)\oplus KX_i(F) \arrow[r] \arrow[d, "\mu_E\oplus\mu_F"] & KX_i(X) \arrow[r, "\partial_{\mathrm{KX}}"] \arrow[d, "\mu_X"] & KX_{i-1}(E\cap F) \arrow[d, "\mu_{E\cap F}"] \\
K_i(C^*(E\cap F)) \arrow[r] & K_i(C^*(E))\oplus K_i(C^*(F)) \arrow[r] & K_i(C^*(X)) \arrow[r, "\partial_{\mathrm{Roe}}"] & K_{i-1}(C^*(E\cap F)).
\end{tikzcd}$$

The vertical maps are the coarse Baum–Connes assembly maps.

Let $Z\subseteq X$ be a net and put

$$Z_E := Z\cap E, \qquad Z_F := Z\cap F.$$

In general,

$$P_r(Z)\neq P_r(Z_E)\cup P_r(Z_F).$$

Indeed, a simplex in $P_r(Z)$ may have all vertices of mutual Z-diameter at most r, with some vertices lying in $E\setminus F$ and others lying in $F\setminus E$.

So we thicken the vertex sets. For $R>r$, define

$$Z_E^{(R)} := Z\cap N_R(E), \qquad Z_F^{(R)} := Z\cap N_R(F),$$

where $N_R(\cdot)$ denotes the R-neighbourhood. Then

$$P_r(Z) = P_r(Z_E^{(R)})\cup P_r(Z_F^{(R)}).$$

To see this, let

$$x=\sum_{z\in Z} t_z z\in P_r(Z).$$

Choose a vertex $z_0\in \operatorname{supp}(x)$. Since $X=E\cup F$, either $z_0\in E$ or $z_0\in F$. If $z_0\in E$, then every vertex $z\in\operatorname{supp}(x)$ satisfies

$$d(z,E)\le d(z,z_0)\le r<R.$$

Thus the whole support of x lies in $Z_E^{(R)}$, and so

$$x\in P_r(Z_E^{(R)}).$$

The case $z_0\in F$ is the same.

The vertex intersection is

$$Z_E^{(R)}\cap Z_F^{(R)} = Z\cap N_R(E)\cap N_R(F).$$

By coarse excisiveness, for this R there is $S>0$ such that

$$N_R(E)\cap N_R(F)\subseteq N_S(E\cap F).$$

Therefore

$$Z_E^{(R)}\cap Z_F^{(R)}\subseteq Z\cap N_S(E\cap F).$$

So the fixed-scale intersection is coarsely controlled by $E \cap F$; conversely, bounded neighbourhoods of $E \cap F$ are coarsely equivalent to $E \cap F$. After passing to direct limits over Rips scales, the intersection contribution is the coarse K-homology group

$$KX_*(E \cap F).$$

Thus the commutative Mayer–Vietoris diagram gives, for every

$$b \in KX_i(X),$$

the identity

$$\partial_{\mathrm{Roe}}(\mu_X(b)) = \mu_{E\cap F}(\partial_{\mathrm{KX}}(b)),$$

where

$$\partial_{\mathrm{KX}} : KX_i(X) \longrightarrow KX_{i-1}(E \cap F)$$

is the Mayer–Vietoris boundary map. Equivalently, if μ_X is an isomorphism and

$$a \in K_i(C^*(X)),$$

then

$$\partial_{\mathrm{Roe}}(a) = \mu_{E\cap F}(\partial_{\mathrm{KX}}(\mu_X^{-1}(a))).$$

Since the indexing is 2-periodic, we identify KX_{i-1} with KX_{i+1} when convenient.

References

[WY20] Rufus Willett and Guoliang Yu, *Higher Index Theory*, Cambridge Studies in Advanced Mathematics, vol. 189, Cambridge University Press, 2020.